

WASTE WATER TREATMENT

*Minor project-1 report submitted
in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology
in
Computer Science & Engineering**

By

AVB RUVANTHIKA	(22UECM0229)	(VTU24145)
CH NAAVYA	(22UECM0053)	(VTU22704)
SHUBHAM KUMAR JHA	(22UECM2019)	(VTU27216)

*Under the guidance of
Dr.J.VIMALA ITHAYAN, M.E., Ph.D.,
Assistant professor*



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE & TECHNOLOGY**

(Deemed to be University Estd u/s 3 of UGC Act, 1956)

**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

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CERTIFICATE

It is certified that the work contained in the project report titled "WASTE WATER TREATMENT" by "AVB RUVANTHIKA (22UECM0229), CH NAAVYA (22UECM0053), SHUBHAM KUMAR JHA (22UECM2019)" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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October, 2024

DECLARATION

We declare that this written submission represents my ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Date: / /

APPROVAL SHEET

This project report entitled WASTE WATER TREATMENT by AVB RUVANTHIKA (22UECM0229), CH NAAVYA (22UECM0053), SHUBHAM KUMAR JHA (22UECM2019) is approved for the degree of B.Tech in Computer Science & Engineering.

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Supervisor

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Date: / /

Place:

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ABSTRACT

This project presents a smart wastewater treatment system that monitors rainwater quality in real-time using sensors for pH, TDS, temperature, and water level. The system analyzes these parameters via an ESP module and AI, determining water quality and sending data to a mobile device for instant user alerts. The sensors assess acidity, purity, temperature, and volume of rainwater, ensuring suitability for use or storage. This automated approach provides accurate, sustainable water management, empowering users to make informed decisions on rainwater use. Real-time alerts enhance quick response, making it an ideal solution for efficient rainwater resource management. This system offers practical benefits, especially in areas prone to water scarcity, by supporting sustainable water management. By providing accurate, real-time insights, users can make informed decisions about rainwater use, storage, and treatment, reducing reliance on traditional water sources. Additionally, AI-based analysis allows for efficient, automated alerts when water quality risks are detected. Testing shows the system's efficacy in maintaining high-quality monitoring standards, with mobile alerts facilitating quick user responses. This comprehensive, automated approach makes the system suitable for both urban and rural environments, encouraging responsible water management practices and addressing the growing need for sustainable water resources.

Keywords

Smart wastewater treatment, rainwater quality monitoring, real-time water quality assessment, pH sensor, TDS sensor, ESP module, AI for water management, mobile alerts, sustainable water management, wireless water monitoring.

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LIST OF ACRONYMS AND ABBREVIATIONS

AI	Artificial Intelligence
CD	Compact Disc
DHT	Digital Humidity and Temperature (sensor)
ESP	Encapsulated Software Protocol
IOS	iPhone Operating System
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
PH	Potential of Hydrogen
SMS	Short Message Service
TDS	Total Dissolved Solids
TLS	Transport Layer Security

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Chapter 1

INTRODUCTION

1.1 Introduction

This project addresses these issues by developing a smart wastewater treatment system to collect and monitor rainwater quality in real time. Utilizing advanced sensors including pH, water level, Total Dissolved Solids (TDS), and temperature sensors the system offers a comprehensive assessment of rainwater. The core of the system is an ESP module that processes the collected data. The TDS sensor gauges dissolved solids indicating water purity, and the water temperature sensor monitors temperature, impacting quality and usability.

Artificial Intelligence (AI) algorithms analyze this data to determine water quality, classifying it as safe or unsafe. This information is wirelessly transmitted to users' mobile devices, providing instant updates on key parameters like temperature, pH levels, and water levels. This real-time monitoring enables effective rainwater management. The system is designed to be scalable, making it suitable for residential, agricultural, and industrial applications. In conclusion, this project represents a significant advancement in water management. By leveraging advanced sensors, AI, and wireless communication, it offers an efficient, scalable solution to ensure safe and usable rainwater, supporting sustainable practices and contributing to environmental conservation.

1.2 Aim of the project

The aim of this project is to develop a smart wastewater treatment system that collects and monitors rainwater quality in real-time. The system utilizes a range of sensors to measure key water properties such as pH, temperature, Total Dissolved Solids, and water levels. By integrating Artificial Intelligence (AI) for water quality assessment and leveraging wireless communication through an ESP module, This enables effective and automated water quality monitoring and management, promoting sustainable water use for residential, agricultural, and industrial applications.

1.3 Project Domain

The domain of this project falls under Environmental Engineering and Smart Technology Integration. Environmental engineering focuses on developing sustainable solutions to manage natural resources effectively, address pollution, and mitigate the impacts of climate change. This project contributes to environmental engineering by creating a system that not only collects and treats rainwater but also ensures its quality through continuous monitoring. By employing advanced sensors and AI technology, the project aligns with the principles of environmental engineering, aiming to provide a sustainable method for water resource management.

Additionally, the project is deeply rooted in the Internet of Things (IoT) and Artificial Intelligence (AI) domains. This integration allows for remote monitoring and management of water quality via mobile devices, showcasing the practical applications of IoT in environmental monitoring. This fusion of IoT and AI within the environmental engineering domain exemplifies how modern technology can be leveraged to address traditional environmental challenges, promoting smarter and more sustainable practices.

1.4 Scope of the Project

The scope of this project encompasses the automation of rainwater collection and quality monitoring to support sustainable water management practices. By employing a variety of sensors, including pH, water level, TDS, and temperature sensors, the system continuously assesses the quality of collected rainwater. The integration of an ESP module and Artificial Intelligence (AI) ensures that the data is processed accurately and efficiently. The system provides real-time feedback to users through mobile devices, enabling them to make informed decisions about water usage.

The ability to remotely monitor and manage water quality via wireless communication also opens up possibilities for implementation in remote or hard-to-access areas. By supporting real-time data access and automated alerts, the system ensures that water management practices are both efficient and sustainable, contributing to broader environmental conservation efforts and addressing the global challenge of water scarcity.

Chapter 2

LITERATURE REVIEW

2.1 Literature Review

Sl.No	Author's Name	Paper name and publication details	Year of publication	Main content of the paper
1	Zhang et al	"Application of Artificial Neural Networks for Predicting Effluent Quality in Wastewater Treatment" - IEEE Environmental Science Journal	2020	Demonstrates the use of Artificial Neural Networks (ANN) for real-time monitoring and control of effluent quality in wastewater treatment systems.
2	Li et al	"Support Vector Machine for Classifying Wastewater Contamination Levels" - IEEE Journal of Water Process Engineering	2019	Explores the use of Support Vector Machines (SVM) for classifying wastewater contamination levels.
3	Wang et al	"Optimization of Aeration Control in Wastewater Treatment Using Genetic Algorithms" - IEEE Journal of Cleaner Production	2021	Describes the use of genetic algorithms to optimize aeration control in wastewater treatment.
4	Singh and Rao	"Time-Series Analysis for Predictive Maintenance in Wastewater Treatment Plants" - IEEE International Journal of Science	2020	Applies time-series analysis for predicting equipment failures in wastewater treatment plants.

Table 2.1: Literature Review

The above table 2.1 describes the Literature review of advanced computational methods.

2.2 Gap Identification

The smart wastewater treatment system for rainwater monitoring offers real-time assessment of water quality through sensors measuring pH, TDS, temperature, and water level, processed by an ESP module with AI. While this design is functional, there are key areas for improvement to enhance accuracy, adaptability, and efficiency. Firstly, the system's scope is limited in contaminant detection. Current sensors capture basic parameters, yet rainwater often contains pollutants like heavy metals and biological contaminants that are crucial for comprehensive quality assessment. Integrating additional sensors could enable a more thorough analysis, making the system suitable for regions with varied contamination risks.

Furthermore, the lack of historical data analysis restricts the system's ability to detect trends or make predictive assessments, which could prevent potential issues. Storing and analyzing historical data would allow for trend-based insights, aiding in proactive maintenance and intervention. Scalability and adaptability are also limited in the current setup, as it is tailored for small-scale rainwater treatment. Developing a modular design could enable expansion to larger facilities or adaptation for diverse water sources, making it useful across various settings.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

In many regions, traditional methods for wastewater treatment and rainwater management involve manual monitoring and periodic water quality testing. These systems typically rely on manual sampling and laboratory analysis to measure key water parameters such as pH, temperature, and total dissolved solids (TDS). While these methods can be accurate, they are labor-intensive, time-consuming, and often not conducted frequently enough to provide real-time data.

This can be particularly problematic during periods of heavy rainfall or pollution events, where rapid changes in water quality can occur. In such cases, the lack of real-time data can lead to ineffective management and delayed corrective actions, which can exacerbate environmental damage and increase public health risks. Additionally, manual systems often lack integration with modern technology, making it difficult to analyze data trends over time or to implement automated responses

3.1.1 Disadvantages of existing system

- **Lack of Real-Time Monitoring:**

Traditional methods rely on periodic manual testing, leading to delays in detecting water quality issues. Absence of continuous data hampers timely decision-making and corrective actions.

- **High Labor and Operational Costs:**

Manual sampling and laboratory analysis require significant time, trained personnel, and specialized equipment. The process is labor-intensive, contributing to higher operational expenses.

- **Increased Potential for Human Error:**

Manual Multiple stages of human intervention increase the likelihood of errors and inconsistencies in data collection and analysis. Data reliability may be compromised, affecting the accuracy of water quality assessments.

- **Delayed Detection and Response:**

Slow identification of contamination or other issues can result in untreated water being released or used. Delayed responses to water quality changes can exacerbate environmental damage and health risks.

3.2 Problem statement

Traditional methods for monitoring and managing wastewater and rainwater are plagued with inefficiencies and limitations. These systems rely heavily on manual sampling and laboratory analysis to measure critical water parameters like pH, temperature, and total dissolved solids (TDS). This process is not only labor-intensive and time-consuming but also fails to provide real-time data, posing significant health risks and environmental hazards. Additionally, the infrequent nature of manual testing makes it difficult to monitor sudden changes in water quality.

The need for trained personnel and specialized laboratory facilities makes these systems expensive to maintain and operate. Making it challenging to implement effective water management strategies. The inability to integrate with modern technologies like artificial intelligence and real-time data analytics limits the potential for predictive maintenance and automated responses, reducing the overall efficiency and responsiveness of these systems.

3.2.1 Advantages of existing system

- **Real-Time Monitoring:**

Continuous data collection provides immediate insights into water quality, enabling prompt detection of contamination and other issues. Instant feedback allows for rapid decision-making and timely intervention.

- **Cost-Effective Operations:**

Automation reduces the need for manual sampling and laboratory analysis, cutting down labor and operational costs. Minimizes the requirement for specialized equipment and trained personnel.

- **Integration with Modern Technology:**

The system leverages AI for predictive analysis, identifying potential issues before they become critical. Seamless integration with data analytics tools allows for advanced trend monitoring and automated responses.

- **Improved Environmental and Public Health:**

Proactive monitoring and management help protect ecosystems and public health by preventing the release or use of contaminated water. Sustainable water management practices contribute to long-term environmental preservation.

- **User-Friendly Interface:**

Real-time updates and alerts delivered to mobile devices ensure users are always informed about water quality. Easy-to-use interface allows for straightforward monitoring and management of water resources.

3.3 System Specification

3.3.1 Hardware Specification

- **ESP32 Microcontroller:**

Dual-core Tensilica LX6 microprocessor, Clock speed up to 240 MHz, 520 KB SRAM, Wi-Fi and Bluetooth connectivity, Multiple GPIO pins for sensor interfacing.

- **pH Sensor:**

High-accuracy pH probe, Measurement range: 0 to 14 pH, Analog output, Waterproof and durable design.

- **TDS (Total Dissolved Solids) Sensor:**

Measurement range: 0 to 1000 ppm, Accuracy: ± 10 , Waterproof probe, Analog output.

- **Water Temperature Sensor (DS18B20):**

Temperature range: -55 to 125°C, Accuracy: $\pm 0.5^\circ\text{C}$, Digital output, Waterproof stainless steel probe.

- **Water Level Sensor:**

Non-contact liquid level sensor, Detection range: 0 to 30 mm, Output: Digital signal, Waterproof and corrosion-resistant.

3.3.2 Software Specification

- **Programming Languages:**

C/C++: Primary language for developing firmware for the ESP32 microcontroller, Python: Used for data analysis and visualization.

- **Microcontroller Firmware:**

ESP-IDF (Espressif IoT Development Framework): Official development framework for ESP32, Arduino Core for ESP32: Simplified programming environment for ESP32.

- **Sensor Libraries:**

Adafruit Unified Sensor Library: Common interface for various sensors, DHT Sensor Library: For interfacing with DHT22 temperature and humidity sensor, pH Sensor Library: Custom or available libraries for pH sensor integration, OneWire and DallasTemperature Libraries: For interfacing with DS18B20 temperature sensor, TDS Sensor Library: Custom or available libraries for TDS sensor integration.]]

- **Mobile Application:**

Blynk Mobile App: Provides a user interface for real-time monitoring and control of the system, Custom Android/iOS App: For additional functionalities and personalized user experience.

- **Cloud and IoT Platforms:**

MQTT (Message Queuing Telemetry Transport): Lightweight messaging protocol for sending data to cloud services, ThingSpeak: IoT analytics platform for aggregating, visualizing, and analyzing live data streams, Blynk: Mobile app for IoT projects, providing real-time monitoring and control.

3.3.3 Standards and Policies

Data Security and Privacy

Encryption: SSL/TLS protocols protect data in transmission, Access Control: Role-based access and multi-factor authentication, Audits: Regular security audits for ongoing risk management.

Standard Used: ISO/IEC 27001

Environmental Management

Energy Efficiency: Power-saving modes to reduce consumption, Sustainable Materials: Eco-friendly hardware selection and e-waste recycling, Regulatory Compliance: Adherence to environmental laws for responsible resource use.

Standard Used: ISO 14001

Software Development and Maintenance

Agile Development: Incremental updates and CI/CD for efficient deployment, Comprehensive Documentation: User manuals and maintenance guides, Code Quality: Automated testing and code reviews for reliability.

Standard Used: ISO/IEC 12207

Chapter 4

METHODOLOGY

4.1 Proposed System

The proposed smart wastewater treatment system is designed to monitor and manage rainwater quality effectively through real-time data collection and advanced analysis. Utilizing a combination of sensors—pH, TDS, water temperature, water level, and DHT (temperature and humidity)—the system provides a comprehensive view of water quality, addressing both basic parameters and specific contaminants.

An ESP (Wi-Fi-enabled microcontroller) module serves as the system's processing unit, gathering sensor data and employing AI algorithms to assess water quality on a nuanced scale. This AI component allows for trend detection and predictive analytics, empowering users with alerts and maintenance recommendations based on historical data trends.

To enhance functionality, the system includes several innovative features. It expands contaminant detection by integrating sensors for specific pollutants, such as heavy metals or microbial contaminants, providing a more thorough assessment suitable for varied environmental conditions. Historical data storage enables trend analysis and predictive assessments, helping users identify emerging issues and address them proactively. The modular design supports scalability, allowing the system to adapt to larger or more complex water treatment setups.

4.2 General Architecture



Figure 4.1: **General Architecture**

The figure 4.1 illustrates the process of a smart wastewater treatment system for assessing rainwater quality. It begins with rainwater collection, where water is gathered for analysis. Following this, sensors measure various water quality parameters, including pH, TDS, and temperature, and collect relevant data. The collected data then undergoes processing, likely handled by an ESP module, to prepare it for further analysis. Next, AI algorithms assess the water quality based on the processed sensor data, determining its condition. The results of this assessment are then communicated, possibly through wireless means, allowing the information to be transmitted to relevant devices. Finally, the data reaches the user interface, where real-time information on water quality is displayed.

4.3 Design Phase

4.3.1 Data Flow Diagram

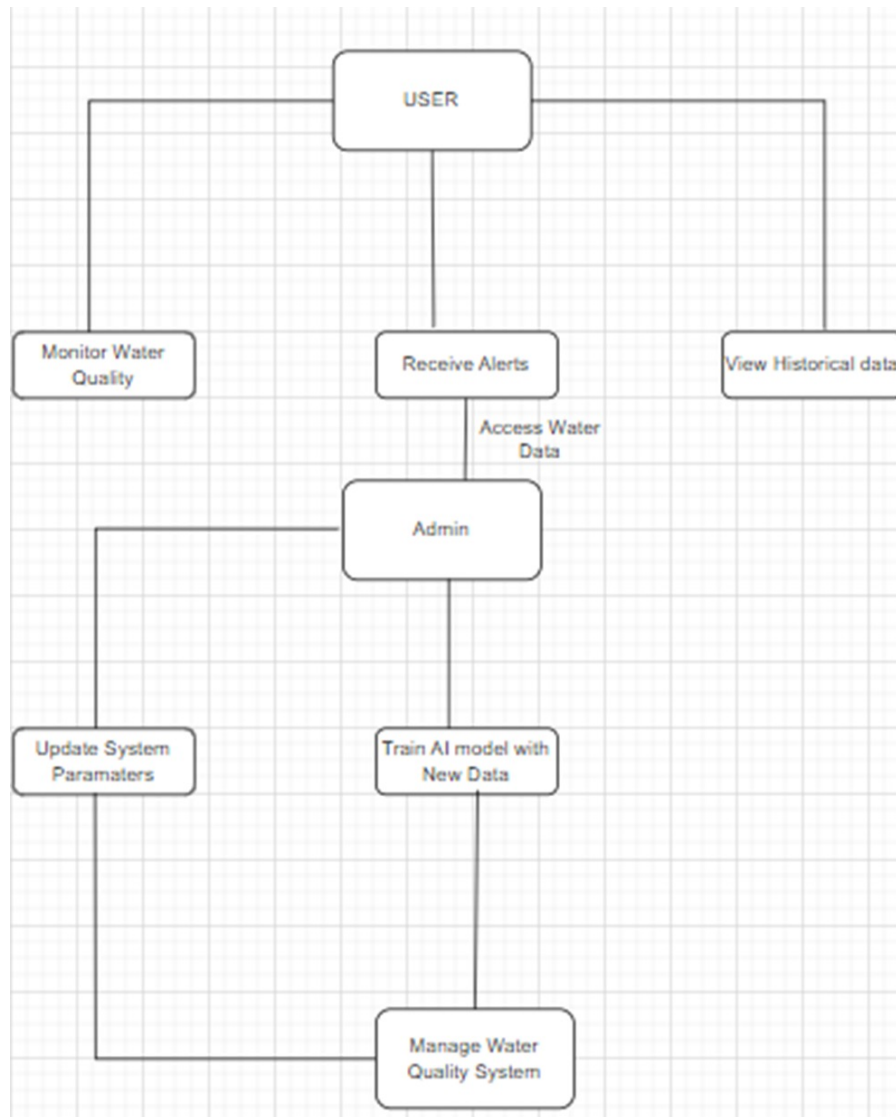


Figure 4.2: **Data Flow Diagram**

The figure 4.2 illustrates Sensors in the water monitor key parameters like pH, turbidity, and temperature in real-time, with the data sent to a central system for comparison against predefined safe thresholds. If the sensor data exceeds these thresholds, an alert is triggered, and notifications are sent to users via SMS, email, and mobile app alerts. Users then receive the alert, review the data, and take corrective action as needed.

4.3.2 Use Case Diagram

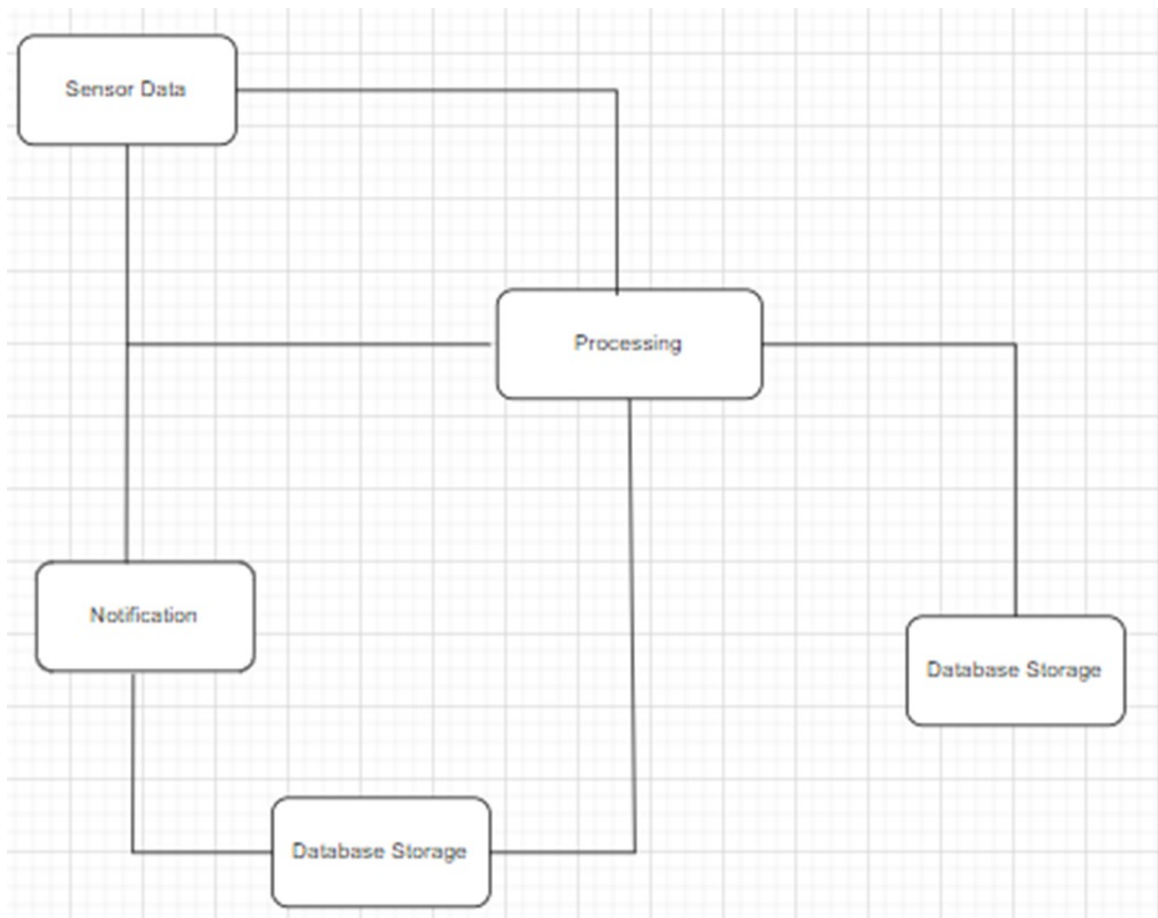


Figure 4.3: Use Case Diagram

The figure 4.3 illustrates Users receive real-time data on water parameters like pH, temperature, and turbidity, and are notified via SMS, email, or app notifications when parameters exceed safe thresholds. They can access past water quality records and trends, while admins have the ability to modify threshold values and system settings. Admins can also improve the AI by uploading new data to optimize predictions.

4.3.3 Class Diagram

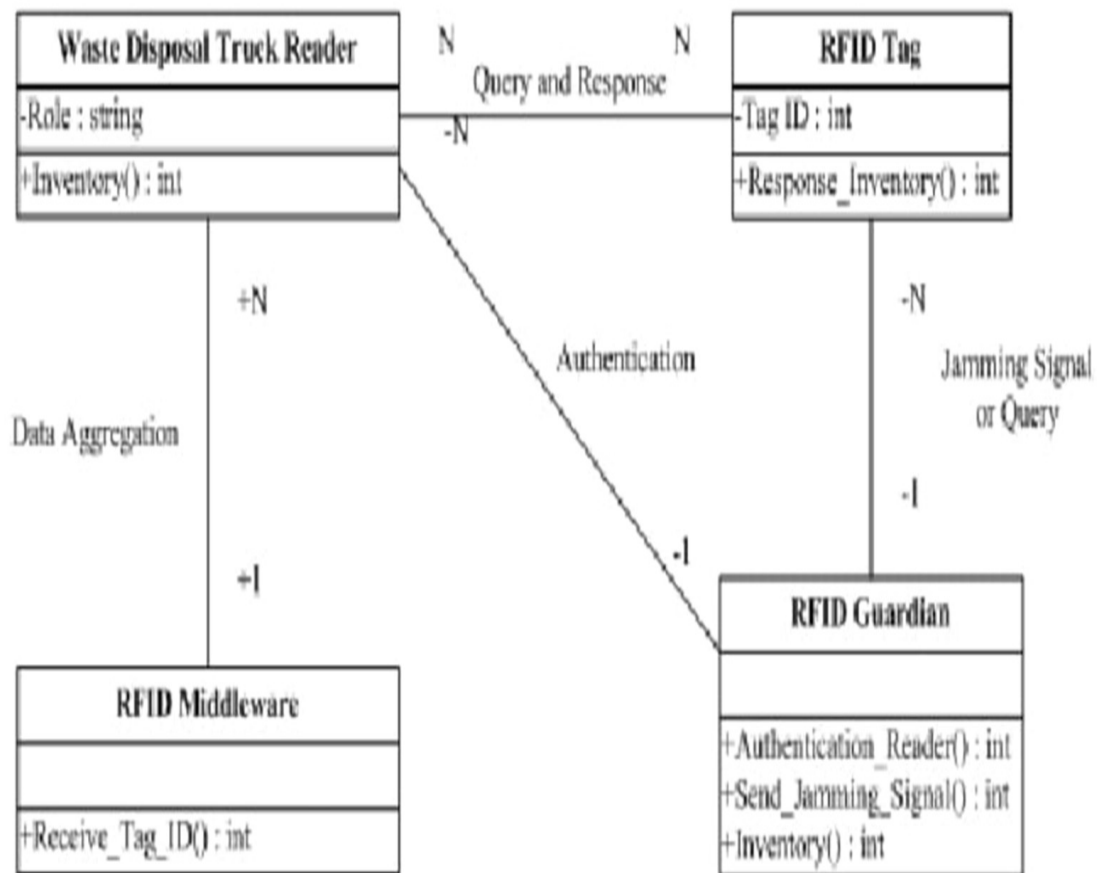


Figure 4.4: **Class Diagram**

The figure 4.4 shows The Class Diagram of water quality monitoring system includes Sensors that gather real-time data and send it to the DataProcessor for analysis against safe thresholds. When thresholds are exceeded, an Alert is triggered and sent via the NotificationSystem (, email, app) to Users and Admins. Users receive alerts and view historical data, while Admins can adjust thresholds, update system settings, and improve the AI Model by uploading new data for better predictions. The AI Model refines anomaly predictions, enhancing system responsiveness. This structure ensures timely alerts and system adaptability to changing water quality conditions.

4.3.4 Sequence Diagram

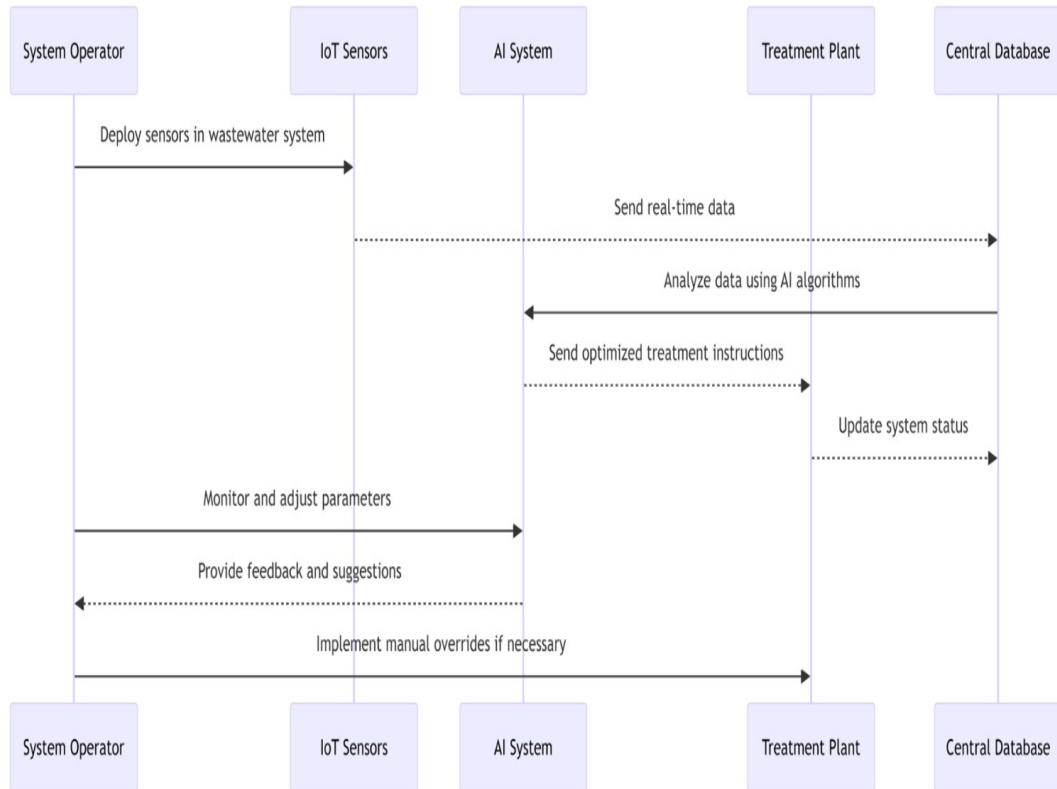


Figure 4.5: Sequence Diagram

The figure 4.5 illustrates Sequence Diagram of The DataProcessor evaluates this data against established thresholds. If any readings exceed these limits, it triggers an Alert that is sent to the NotificationSystem. This system then notifies Users through SMS, email, or app alerts. Upon receiving these notifications, users can review the data and take necessary actions. Additionally, Admins can adjust thresholds within the DataProcessor and retrain the AI Model using new data, improving overall accuracy and responsiveness.

4.3.5 Collaboration diagram

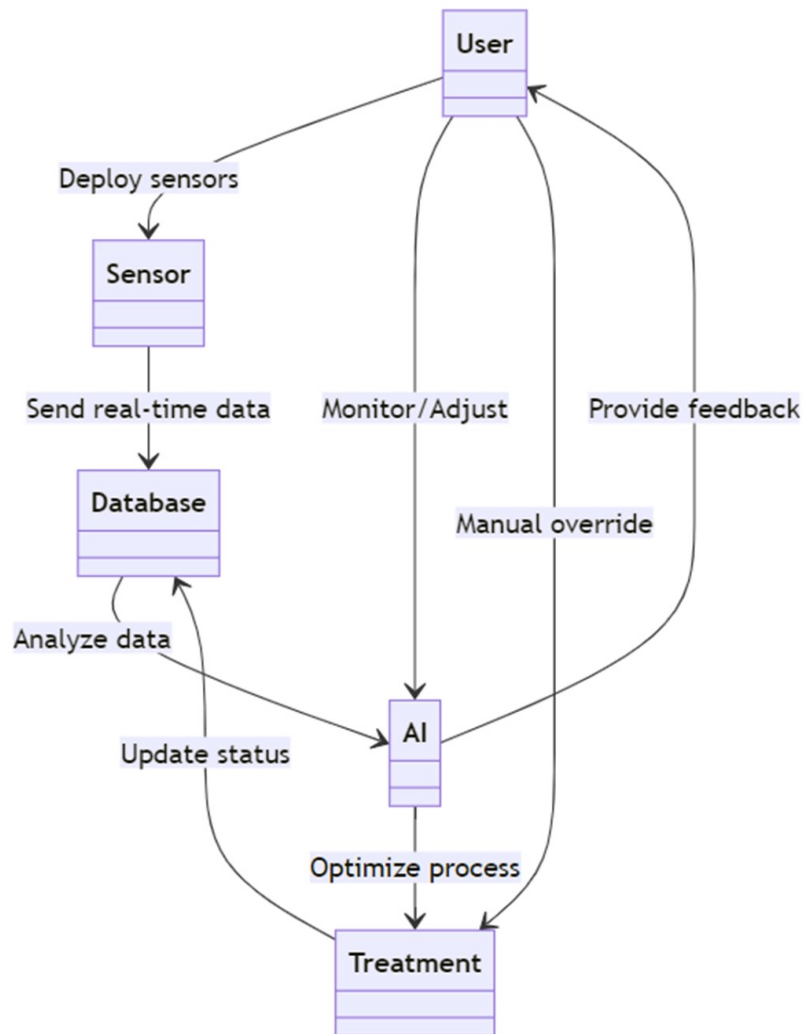


Figure 4.6: Collaboration diagram

The figure 4.6 shows A collaboration diagram for a wastewater treatment system that illustrates the interactions between key components such as the Wastewater Inlet, Pre-treatment Unit, Primary Treatment Unit, Secondary Treatment Unit, Tertiary Treatment Unit, and Output/Discharge. Raw wastewater enters through the inlet and is sent to the pre-treatment unit, where solids are removed. It then moves to the primary unit for sedimentation, followed by biological treatment in the secondary unit. After tertiary treatment, the purified water is discharged safely. Throughout the process, sensors monitor water quality, allowing for real-time adjustments, while operators ensure compliance with environmental regulations and operational standards.

4.3.6 Activity Diagram

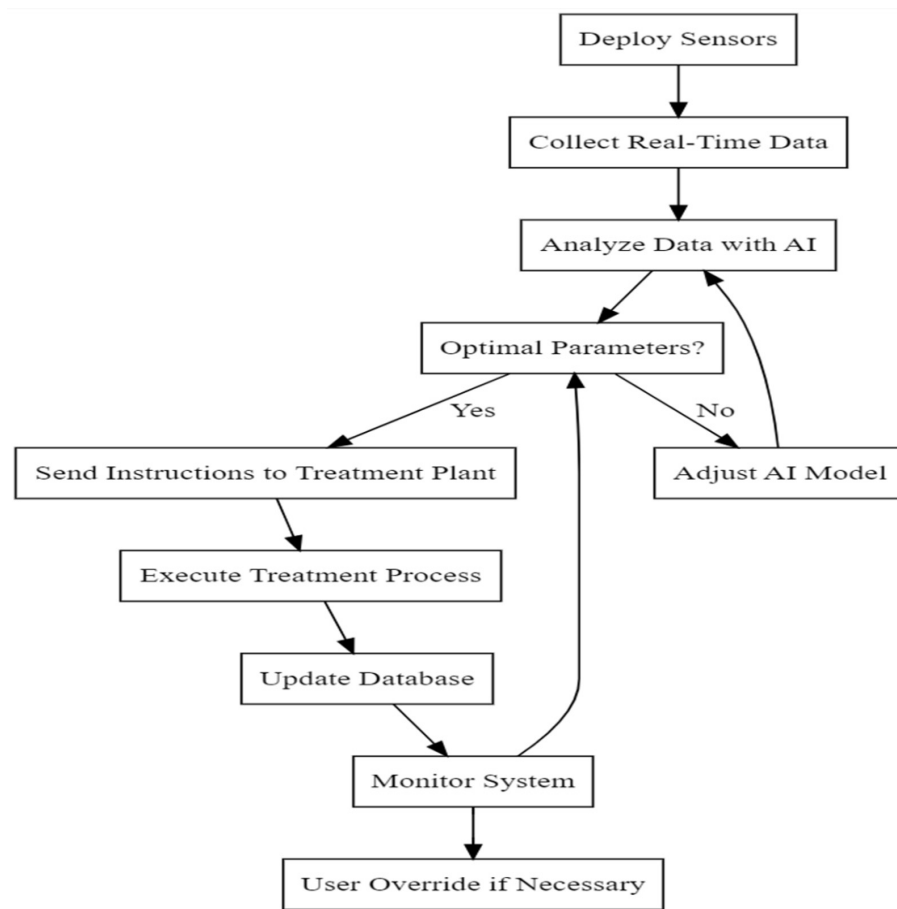


Figure 4.7: Activity Diagram

The figure 4.7 illustrates An activity diagram for wastewater treatment outlines the flow of processes involved in treating wastewater. It begins with the raw wastewater inflow, which first undergoes screening to remove large debris. Next, the wastewater enters the sedimentation tank for primary treatment, allowing solids to settle. The clarified water then moves to the biological treatment stage, where microorganisms degrade organic matter. After this, the water undergoes filtration and disinfection during tertiary treatment. Finally, the treated water is either discharged into the environment or reused. Throughout the process, monitoring occurs to ensure compliance with safety and environmental standards.

4.4 Algorithm & Pseudo Code

4.4.1 Algorithm

The high-level algorithm for a wastewater treatment monitoring system that integrates data collection, analysis, alert generation, and user notifications. This algorithm outlines the main processes involved in monitoring wastewater treatment, ensuring that any deviations from safe conditions are promptly addressed while allowing for user and admin engagement.

1. Initialize System

- Set up sensors for pH, turbidity, temperature, and other relevant parameters.
- Define safe threshold values for each parameter.

2. Data Collection

- Continuously collect data from sensors at predefined intervals.
- Store collected data in a database.

3. Data Analysis

- Retrieve the latest data from the database.
- For each parameter, compare the current values against the predefined thresholds.

4. Check for Threshold Breaches

- If any parameter exceeds its safe threshold
- Trigger an alert.

5. Generate Alert

- Create an alert message containing:
 - Affected parameter(s)
 - Current values
 - Timestamp of the reading
- Store the alert in the alert log.

6. Send Notification

- Use the notification system to send alerts to users via:

- SMS
 - Email
 - Mobile app notifications
- Ensure users receive the alerts based on their preferences.

7.User Interaction

- Allow users to view real-time data and historical records through the application.
- Provide options for users to acknowledge alerts and take corrective actions.

8.Admin Functions

- Provide an interface for admins to:
 - Update threshold values.
 - Upload new data for AI model training.
 - Review system performance and alerts.

9. AI Model Training (Optional)

- Periodically retrain the AI model using collected data to improve prediction accuracy and anomaly detection.

10.Repeat Process

- Continuously monitor the system, repeating steps 2 to 9 as needed.

4.4.2 Pseudo Code

This pseudocode for a wastewater treatment monitoring system that covers data collection, threshold monitoring, alerting, and user interactions:

```
1 SET safeThresholds = { pH: 6.5–8.5, turbidity: max 5 NTU, temperature: max 35 C }
2 INITIALIZE sensors for pH, turbidity, temperature
3 INITIALIZE notification channels (SMS, Email, App)
4
5 WHILE system is running:
6     // Step 1: Data Collection
7     FOR each sensor IN sensors:
8         currentData[sensor] = sensor.readData()
9     END FOR
10
11    // Step 2: Data Analysis
12    FOR each parameter IN currentData:
13        IF currentData[parameter] > safeThresholds[parameter]:
14            // Step 3: Trigger Alert
15            alertMessage = "ALERT: " + parameter + " exceeds threshold. Current value: " +
                currentData[parameter]
16            recordAlert(alertMessage)
17
18            // Step 4: Send Notification
19            SEND alertMessage via SMS, Email, and App to users
20        END IF
21    END FOR
22
23    FOR each user IN users:
24        IF user receives alert:
25            user.viewAlert(alertMessage)
26            user.takeCorrectiveAction()
27        END IF
28    END FOR
29
30    FOR each admin IN admins:
31        IF admin updates threshold OR uploads new data:
32            updateThresholds()
33            IF new data uploaded:
34                trainAIModel()
35            END IF
36        END IF
37    END FOR
38
39    // Repeat data collection and monitoring
40    WAIT for predefined interval
41 END WHILE
```

4.4.3 Data Set / Generation of Data

In a wastewater treatment monitoring system, data generation involves continuously collecting readings from various sensors placed throughout the treatment process. These sensors measure key parameters that indicate water quality, such as pH, turbidity, Temperature, dissolved oxygen, chemical oxygen demand (COD), and biological oxygen demand (BOD). Each sensor produces real-time data streams that are timestamped and tagged with specific locations within the treatment facility.

Data is typically recorded at regular intervals (e.g., every minute or hour) and stored in a central database. This data collection can be enhanced through simulated datasets, especially during system testing or when historical data is unavailable. Simulated data can emulate different scenarios like regular operations, threshold breaches, and system malfunctions, allowing for robust model training and alert testing. This dataset becomes the basis for analyzing water quality trends, assessing system performance, and enabling predictive maintenance.

4.5 Module Description

4.5.1 Module1:Data Collection Module

This module includes a network of sensors pH, TDS, temperature, water level, and DHT sensors that continuously collect data on rainwater quality. Each sensor measures specific parameters such as acidity, dissolved solids, temperature, water volume, and environmental conditions (humidity and air temperature). This data is sent to the ESP module for further processing.

4.5.2 Module2:Data Processing and AI Analysis Module

The ESP module serves as the central processing unit, gathering raw data from the sensors. It uses AI algorithms to analyze this data against predefined thresholds to determine water quality. The AI assesses multiple parameters in real-time, flagging anomalies and identifying water that may be unsafe for use or storage.

4.5.3 Module3:Alert and Notification Module

If the analysis module detects that any parameter exceeds safe limits, this module triggers an alert. Notifications are sent wirelessly via the ESP module to the user's mobile device, ensuring they are immediately informed. The alerts include details on the specific parameter that triggered the alert, allowing for quick action.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

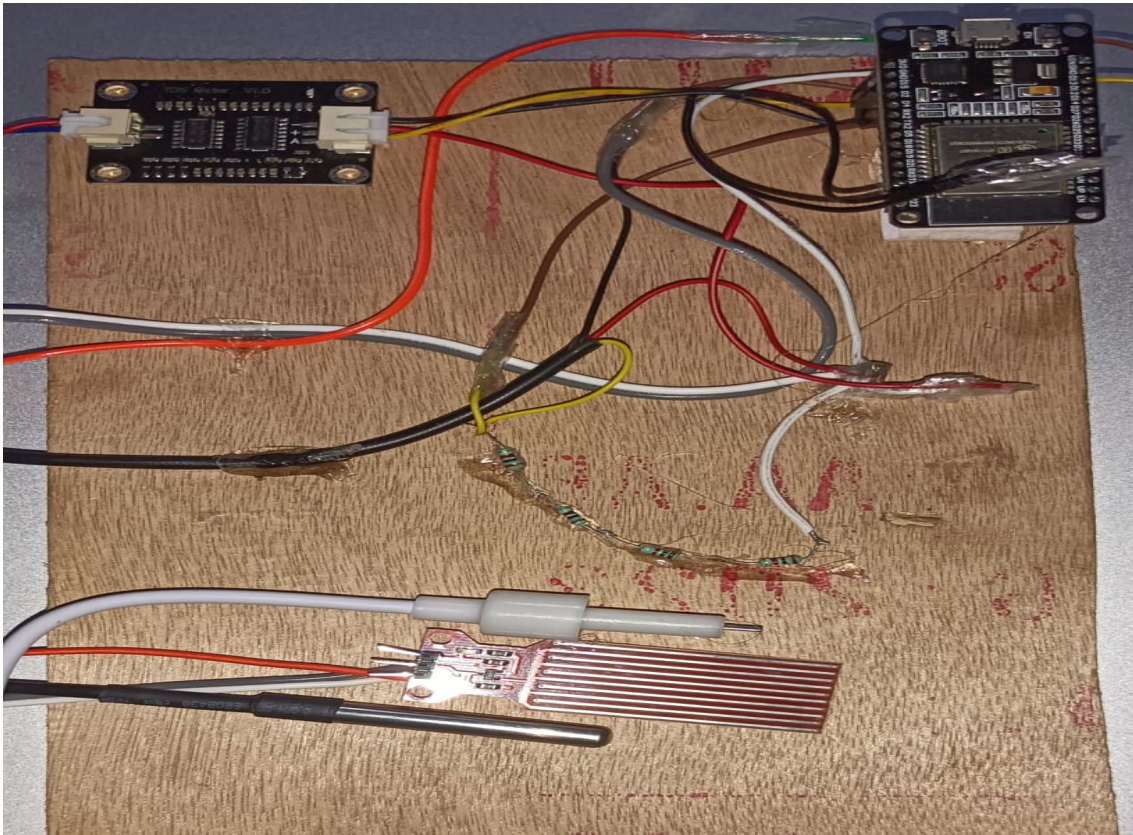


Figure 5.1: Input Design

The figure 5.1 Shows the inputs from various sensors, including a pH sensor, TDS sensor, temperature sensor, and water level sensor, to monitor water quality. These sensors provide raw data on water purity, temperature, and volume. The ESP32 processes this data to determine water usability and sends it to a mobile app for real-time feedback.

5.1.2 Output Design

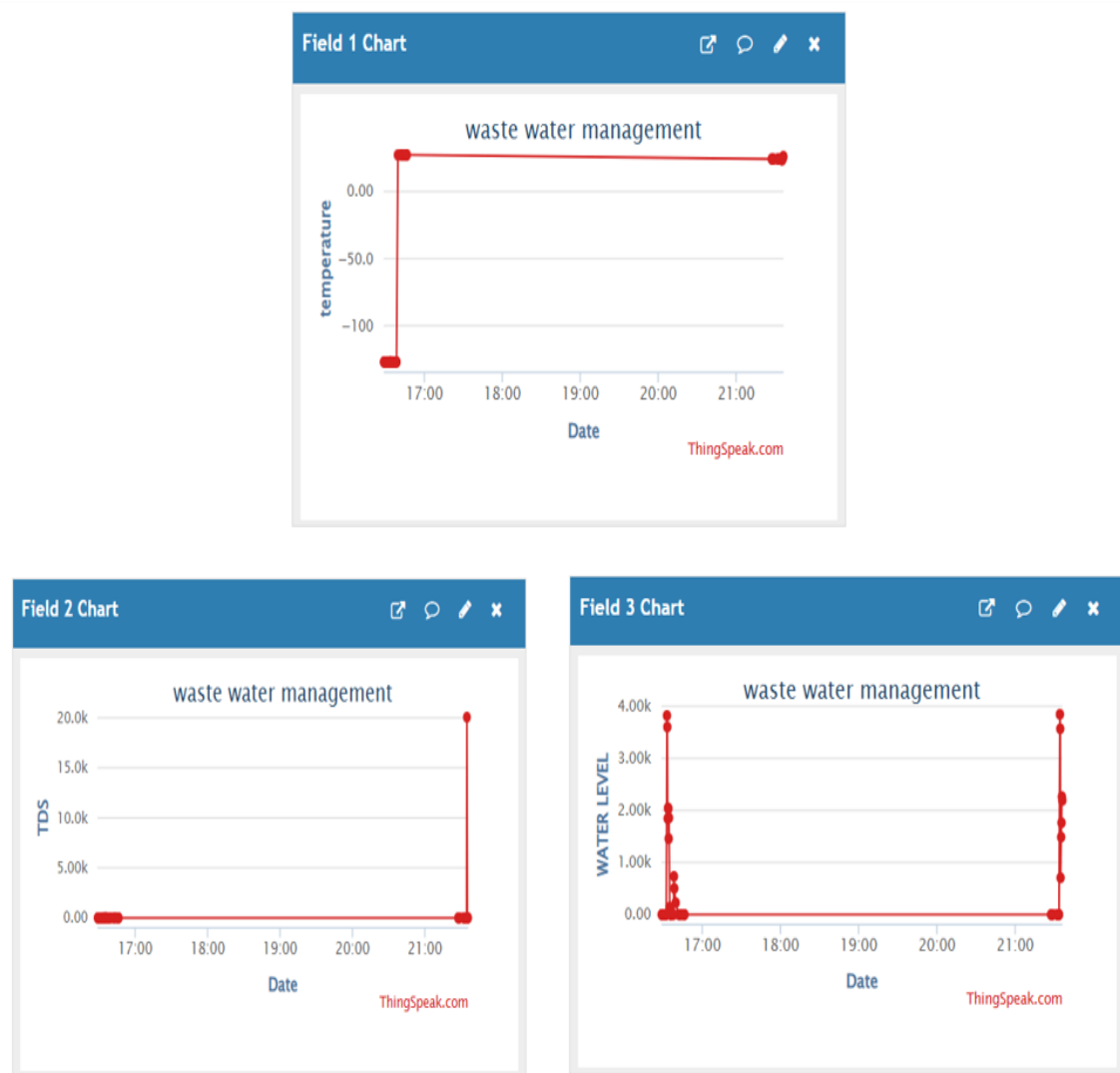


Figure 5.2: **Output Design**

The figure 5.2 illustrates The outputs real-time water quality metrics, including TDS, temperature, and water level, displayed on app. Alerts are generated if parameters exceed safe thresholds, ensuring immediate user awareness. Additionally, data is sent to ThingSpeak for remote monitoring and analysis, enabling efficient water management.

5.2 Testing

5.2.1 Types of Testing

5.2.2 Unit testing

The unit testing for the smart wastewater monitoring system verified the functionality of individual components, including sensors, data processing, Wi-Fi connectivity, and display. The TDS sensor was tested with a calibration solution and provided accurate readings within ± 10 . The temperature sensor was validated against a reference thermometer, showing results within $\pm 1^\circ\text{C}$. The water level sensor displayed linear accuracy when tested with varying water levels. Data processing tests confirmed the accuracy of TDS and EC calculations, including temperature compensation. Wi-Fi successfully connected to the network, and data was transmitted to ThingSpeak without issues. The display correctly presented TDS, temperature, and water level readings. All tests passed, confirming the system's reliability.

5.2.3 Integration testing

Integration testing for the smart wastewater monitoring system verified that all components worked together seamlessly. Sensor data from the TDS, temperature, and water level sensors were processed accurately, and results were displayed on the screen. The system successfully transmitted processed data to ThingSpeak via Wi-Fi, ensuring real-time monitoring. Interactions between sensors, data processing, and the display module were validated, with no data loss or misrepresentation observed. Communication between the ESP32, Wi-Fi network, and ThingSpeak was consistent, and alerts were triggered correctly when thresholds were exceeded. All integration tests passed, confirming the system's ability to function as a cohesive unit.

5.2.4 System testing

System testing for the smart wastewater monitoring system validated its overall functionality and performance under real-world conditions. The system accurately collected data from TDS, temperature, and water level sensors, processed it reliably, and displayed the results on the screen. Data was transmitted successfully to the ThingSpeak server, ensuring real-time remote monitoring. The system handled multiple simultaneous operations, such as data collection, processing, display, and wireless transmission, without delays or errors. Threshold-based alerts were triggered correctly and sent to the user. All tests, including functional, performance, and usability checks, passed, confirming the system's reliability, efficiency, and readiness for deployment.

5.2.5 Test Result

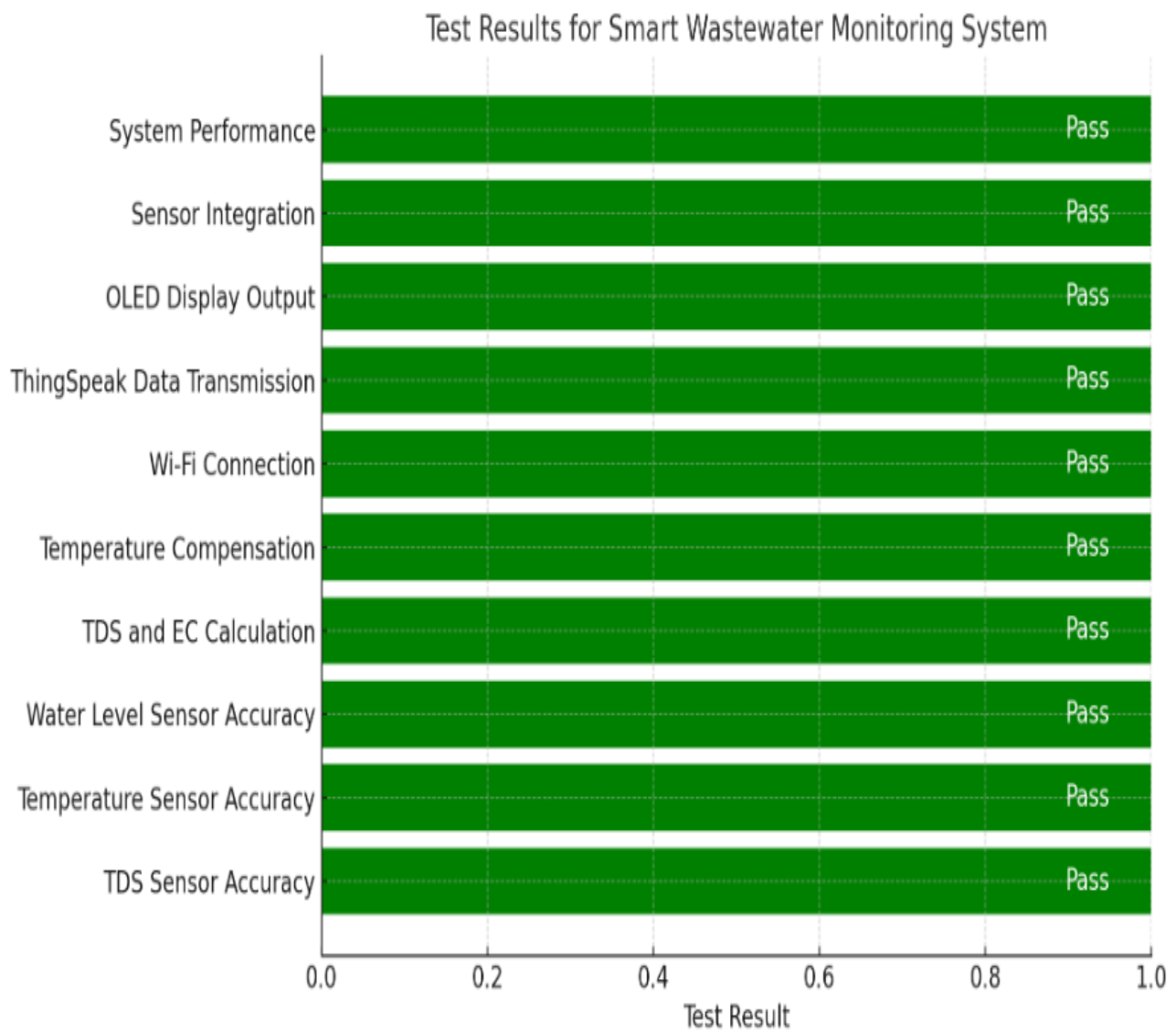


Figure 5.3: Test Image

The figure 5.3 illustrates The system offers real-time and historical data on water temperature, TDS levels, and water storage levels, all accessible via a mobile app. It helps users monitor water quality and availability, providing insights into temperature fluctuations, dissolved impurities, and storage trends. This ensures efficient management and informed decisions for rainwater use or treatment.

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

Integrating AI into wastewater treatment systems significantly enhances operational efficiency and effectiveness. By enabling real-time monitoring and dynamic adjustments, AI ensures that essential water quality parameters—like pH, dissolved oxygen, and contamination levels—are continuously optimized. This responsive monitoring system allows immediate adjustments to dosing, aeration, and other processes based on current water conditions, improving the overall quality of treated water while reducing resource use. Predictive maintenance, another key advantage, helps identify and address equipment issues before they cause downtime, allowing smoother and more consistent plant operations.

In addition to automation, AI's ability to detect anomalies like unexpected contaminant spikes or leaks adds another layer of efficiency. By proactively alerting operators, AI minimizes environmental risks and prevents potential regulatory violations. The cumulative effect of these benefits is a notable reduction in labor, energy, and chemical costs, making AI an economically and environmentally beneficial solution for wastewater treatment. This approach allows treatment plants to meet increasingly stringent standards while maintaining lower operational costs and achieving better sustainability outcomes.

6.2 Comparison of Existing and Proposed System

Existing System: Traditional wastewater treatment systems rely on manual, periodic testing of water quality parameters, often requiring samples to be analyzed in laboratories or on-site, which can delay detection of contaminants. These systems generally operate on fixed schedules, adjusting treatment processes based on pre-set guidelines rather than real-time data. Consequently, changes in water quality are sometimes identified too late, leading to inefficient chemical and energy usage, inconsistent water quality, and limited responsiveness to fluctuating contaminant levels. Maintenance in these systems is often reactive, waiting until equipment fails, resulting in unplanned downtimes and increased repair costs.

Proposed System: The AI-based wastewater treatment system automates monitoring and quality assessment through sensors and real-time data analysis. This system continuously gathers data on key parameters, including pH, temperature, TDS, and water level, and uses AI algorithms to process and assess water quality instantly. When contaminants or quality issues are detected, automated alerts are sent to users' mobile devices, enabling quick response and adjustments. AI also optimizes resource usage by adapting chemical dosing and energy consumption based on current needs, reducing operational costs and environmental impact. With predictive maintenance, this system anticipates equipment issues before they lead to failure, minimizing downtime and extending equipment life. Additionally, the mobile interface allows for real-time data access, historical records, and trend analysis, supporting efficient, sustainable water management.

```

1 START
2
3 INITIALIZE OLED display with dimensions 128x64
4 INITIALIZE OneWire and DallasTemperature for temperature sensor
5 SET pin configurations for TDS sensor, water level sensor, and temperature sensor
6 SET Wi-Fi credentials (SSID and Password)
7 SET ThingSpeak server details and API key
8
9 FUNCTION sendSensorData():
10     REQUEST temperature data from Dallas sensor
11     STORE water temperature in Celsius
12     CALCULATE raw EC value from TDS sensor
13     APPLY temperature coefficient correction for EC
14     CALCULATE TDS value using EC and calibration constants
15     READ water level from water level sensor
16     DISPLAY TDS, EC, temperature, and water level on Serial Monitor
17     WAIT 2 seconds
18
19 FUNCTION sendToThingSpeak():
20     CONNECT to ThingSpeak server
21     IF connection successful:
22         PREPARE HTTP POST data with temperature, TDS, and water level
23         SEND data to ThingSpeak
24         CLOSE connection
25     END IF
26
27 FUNCTION displayData():
28     CLEAR OLED display
29     PRINT TDS value on OLED
30     PRINT EC value on OLED
31     PRINT water temperature on OLED
32     PRINT water level on OLED
33     REFRESH OLED display
34     WAIT 1.5 seconds
35

```



```
36 FUNCTION setup():
37     START Serial Monitor
38     INITIALIZE OLED display
39     CONNECT to Wi-Fi
40     WAIT until Wi-Fi is connected
41     DISPLAY "Wi-Fi Connected" on Serial Monitor
42
43 FUNCTION loop():
44     CALL sendSensorData()
45     CALL displayData()
46     CALL sendToThingSpeak()
47
48 CALL setup()
49
50 REPEAT loop() FOREVER
51
52 END
```

Output

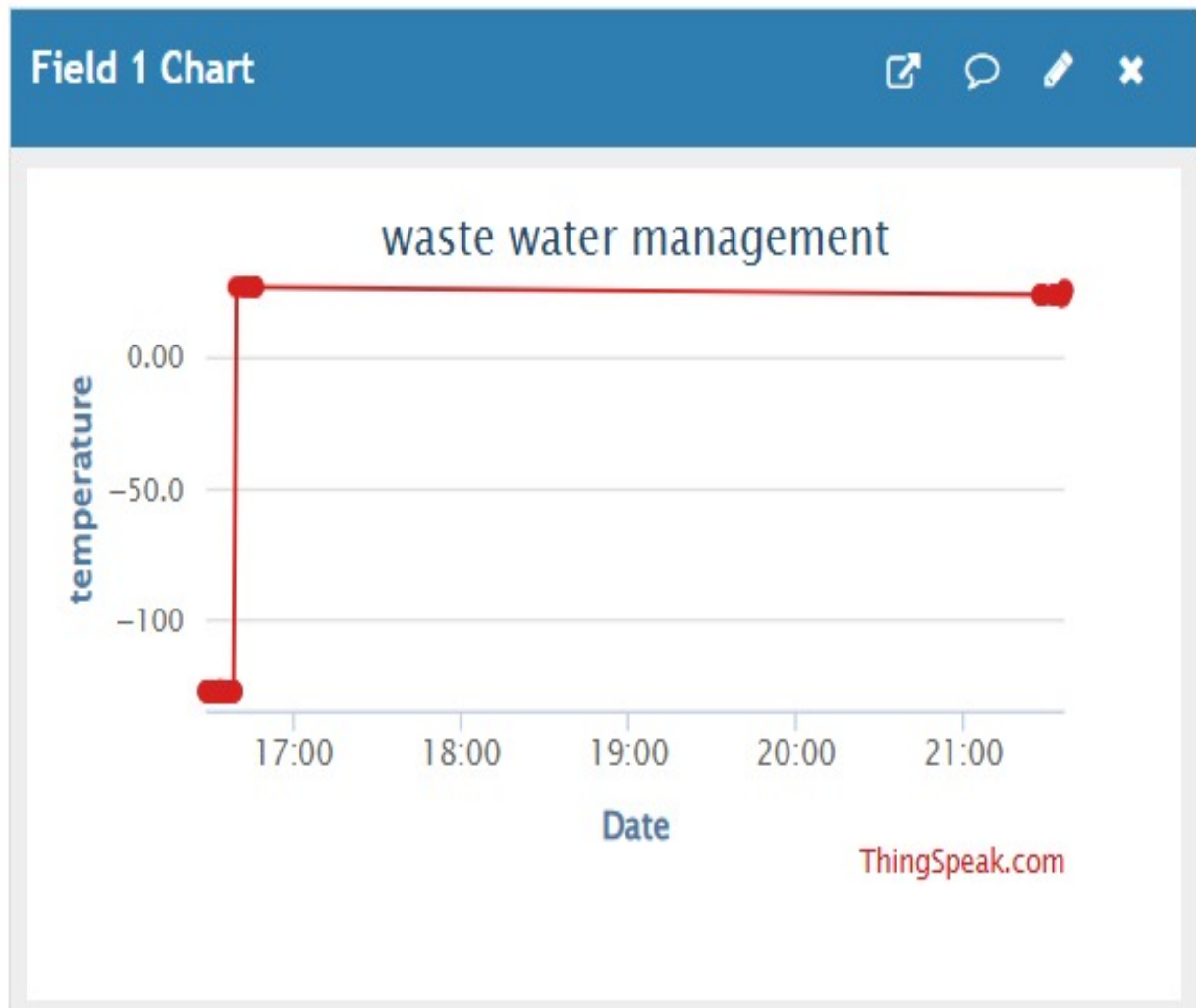


Figure 6.1: **Output 1**

The figure 6.1 shows The Temperature Observation Graph provides real-time and historical data on water temperature, helping users track changes that may impact water quality. This graph enables quick identification of temperature fluctuations over time, which can affect water usability. Accessible on the mobile app, it supports efficient monitoring and management of stored rainwater.



Figure 6.2: **Output 2**

The figure 6.2 shows TDS Observation Graph real-time and historical trends in Total Dissolved Solids (TDS) levels, helping users monitor water purity. This graph provides insight into water quality changes over time, indicating any increase in dissolved impurities. Accessible via the mobile app, it supports informed decisions on rainwater use or treatment needs.

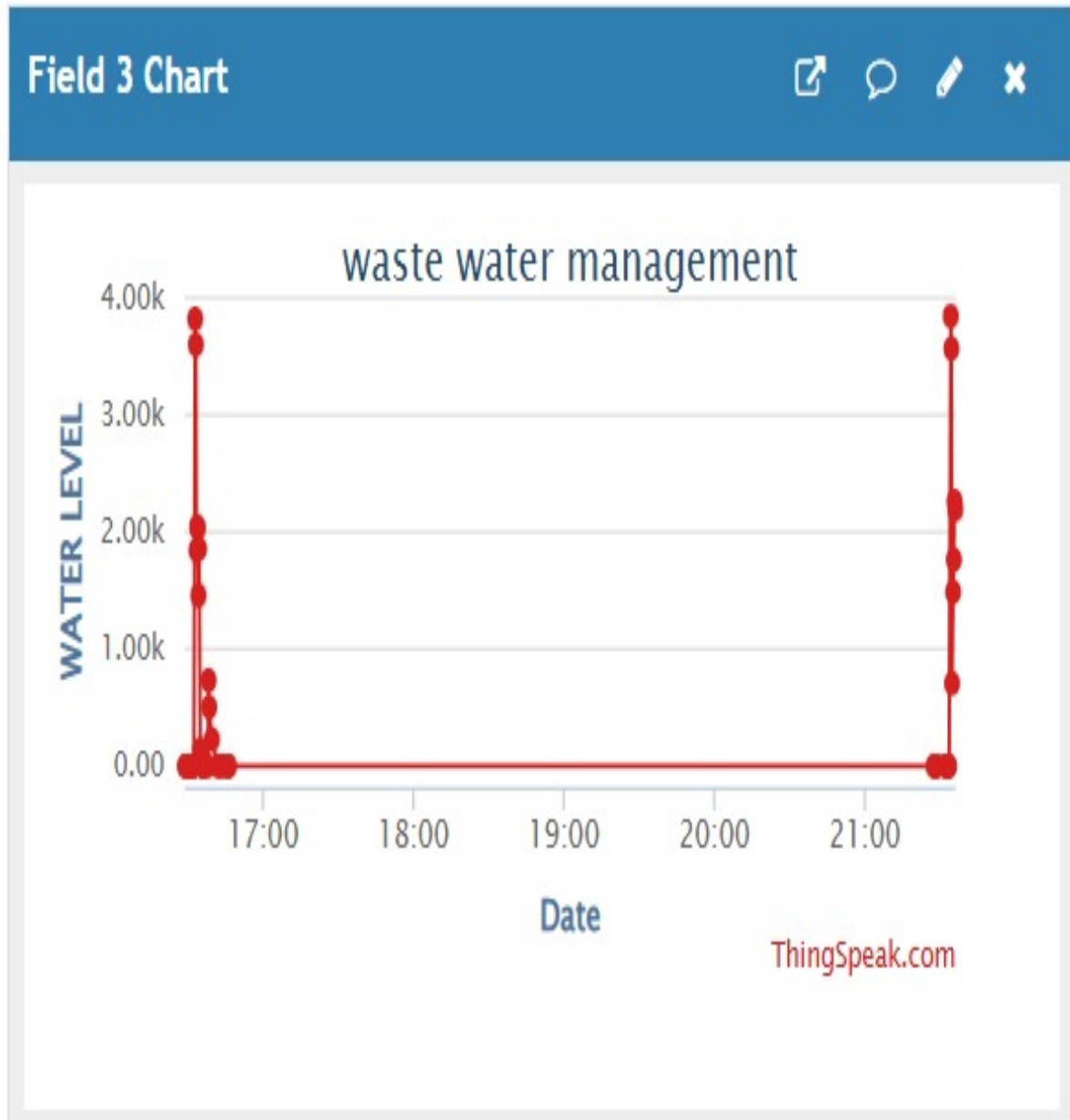


Figure 6.3: **Output 3**

The figure 6.3 shows Water Level Observation Feature displays real-time and historical graphs of water levels, helping users track and manage rainwater storage. This visual data provides insights into collection patterns and storage trends, supporting effective water use planning. Accessible through the mobile app, it allows convenient monitoring of water availability.

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

In conclusion, the development of a smart wastewater treatment system for rainwater collection and quality monitoring represents a significant advancement in sustainable water management practices. This project successfully integrates advanced sensors, an ESP module, and Artificial Intelligence (AI) to provide real-time, automated assessments of rainwater quality. By offering immediate feedback on key parameters such as pH, temperature, TDS, and water levels, the system empowers users to make informed decisions about water usage and treatment. The automated nature of the system reduces the need for manual intervention, ensuring more efficient and timely management of water resources.

Furthermore, the scalability of the system highlights its potential for broader applications across various sectors. From residential water management to agricultural irrigation and industrial processes, the system can be adapted to meet diverse needs. Its capability for remote monitoring via wireless communication ensures that even in remote or hard-to-reach areas, water quality can be effectively managed. By promoting the efficient and sustainable use of rainwater, this project contributes to the global efforts to address water scarcity and environmental conservation. Overall, the smart wastewater treatment system stands as a testament to how modern technology can be harnessed to solve pressing environmental challenges, paving the way for smarter and more sustainable water management solutions.

7.2 Future Enhancements

Looking forward, several enhancements can be made to the smart wastewater treatment system to further improve its functionality and broaden its applications. One potential enhancement is the integration of additional sensors to monitor a wider range of water quality parameters, such as turbidity, microbial content, and chemical pollutants. This would provide a more comprehensive analysis of water quality, ensuring that the collected rainwater meets all necessary safety standards for various uses. Additionally, incorporating solar panels to power the system could make it more sustainable and suitable for deployment in remote areas without reliable access to electricity.

Another area for future enhancement is the development of a more sophisticated AI model capable of predictive analytics. By analyzing historical data, the AI could predict future water quality trends and potential contamination events, allowing for proactive management and timely interventions. This predictive capability could be particularly valuable in agricultural and industrial applications, where anticipating water quality issues could prevent crop damage or process disruptions. Furthermore, enhancing the system's connectivity features, such as integrating it with smart home systems or broader IoT networks, could enable more seamless and user-friendly interactions. This would facilitate better data sharing and decision-making, contributing to more efficient and integrated water management practices.

Overall, these future enhancements would not only improve the system's performance and reliability but also expand its applicability across different sectors and environments. By continuously evolving and incorporating new technologies, the smart wastewater treatment system can remain at the forefront of sustainable water management solutions, addressing emerging challenges and meeting the needs of a diverse range of users.

Chapter 8

PLAGIARISM REPORT

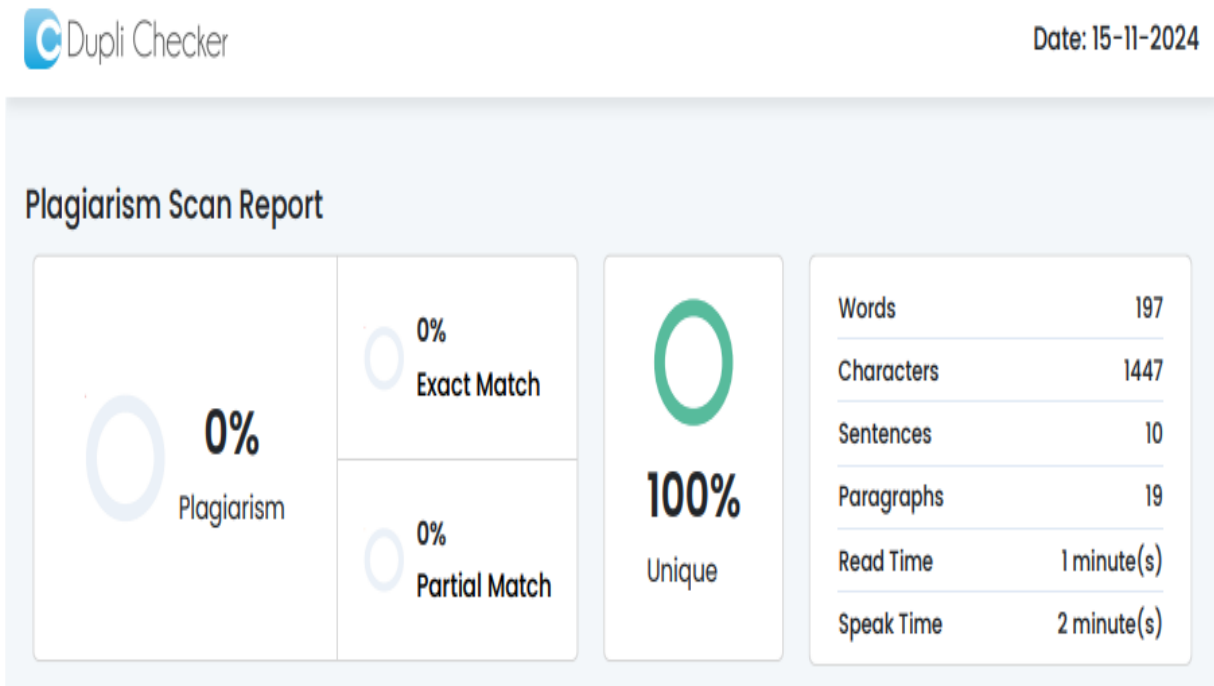


Figure 8.1: Plagiarism Scan Report 1

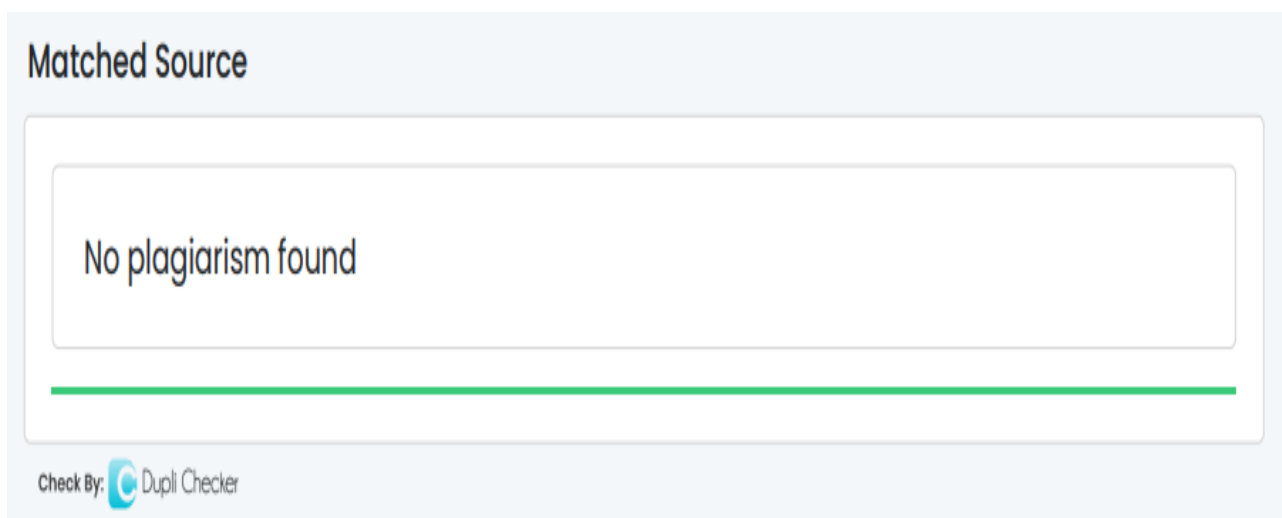


Figure 8.2: Plagiarism Scan Report 2

Appendices

Appendix A

Sample Source Code

```
1 #include <WiFi.h> // ESP32 WiFi library
2 #include <Adafruit_GFX.h>
3 #include <Adafruit_SSD1306.h>
4 #include <DallasTemperature.h>
5 #include <OneWire.h>
6
7 // Pin Definitions
8 #define SCREEN_WIDTH 128 // OLED display width, in pixels
9 #define SCREEN_HEIGHT 64 // OLED display height, in pixels
10 #define OLED_RESET -1 // Reset pin # (or -1 if sharing Arduino reset pin)
11 #define TDS_Sensor 36 // TDS Sensor connected to GPIO36
12 #define TEMP_PIN 13 // OneWire data pin for the temperature sensor
13 #define WATER_LEVEL_PIN 34 // Analog water level sensor connected to GPIO34
14
15 // Create display object
16 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
17
18 // Setup OneWire bus and DallasTemperature
19 OneWire oneWire(TEMP_PIN);
20 DallasTemperature sensors(&oneWire);
21
22 // Constants
23 const float Aref = 3.3; // Reference voltage for TDS
24 const float ecCal = 1.0; // Calibration constant for EC
25 String apiKey = "JPDBS7AW2L6R4D6Z"; // Your ThingSpeak Write API Key
26 const char *ssid = "Naavya"; // WiFi SSID
27 const char *pass = "999999999"; // WiFi Password
28 const char* server = "api.thingSpeak.com"; // ThingSpeak server
29
30 WiFiClient client;
31
32 // Sensor values
33 float ec_Val = 0;
34 unsigned int tds_value = 0;
35 float wTemp = 0;
36 int water_level = 0; // To store water level value
37
38 void sendSensorData() {
39 // Get temperature from Dallas Sensor
40 sensors.requestTemperatures(); // Get temp from all sensors
41 wTemp = sensors.getTempCByIndex(0); // Assuming one sensor
```

```

42
43 // Calculate EC and TDS
44 float V_level = Aref / 1024.0;
45 float rawEc = analogRead(TDS_Sensor) * V_level; // Raw EC data
46 float T_Cof = 1.0 + 0.02 * (wTemp - 25.0); // Temperature Coefficient
47
48 ec_Val = (rawEc / T_Cof) * ecCal; // Temperature and calibration compensation
49 tds_value = (133.42 * pow(ec_Val, 3) - 255.86 * ec_Val * ec_Val + 857.39 * ec_Val) * 0.5;
50
51 // Convert temperature to Fahrenheit
52 double Far = ((wTemp * 9) / 5) + 32;
53
54 // Get water level from the sensor
55 water_level = analogRead(WATER_LEVEL_PIN); // Read water level sensor value (0 to 4095)
56
57 // Output to Serial Monitor
58 Serial.print("TDS: "); Serial.println(tds_value);
59 Serial.print("EC: "); Serial.println(ec_Val, 2);
60 Serial.print("Temperature (C): "); Serial.println(wTemp, 2);
61 Serial.print("Temperature (F): "); Serial.println(Far, 2);
62 Serial.print("Water Level: "); Serial.println(water_level);
63 Serial.println();
64
65 delay(2000); // Delay to avoid rapid repeated readings
66 }
67
68 void setup() {
69     Serial.begin(9600);
70     delay(100);
71
72     // Initialize display
73     display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
74     display.clearDisplay();
75     display.setCursor(25, 25);
76     display.setTextSize(1);
77     display.setTextColor(WHITE);
78     display.println("TDS Sensor");
79     display.setCursor(25, 40);
80     display.setTextSize(1);
81     display.print("Initializing");
82     display.setCursor(30, 5);
83     display.setTextSize(1);
84     display.print("Group P2");
85     display.display();
86     delay(3000);
87
88     // Connect to WiFi
89     Serial.println("Connecting to WiFi...");
90     WiFi.begin(ssid, pass);
91     while (WiFi.status() != WL_CONNECTED) {

```

```

92     delay(500);
93     Serial.print(".");
94 }
95 Serial.println("\nWiFi connected");
96 }
97
98 void loop() {
99     sendSensorData(); // Get sensor data
100
101     // Display data on OLED screen
102     display.clearDisplay();
103     display.setTextSize(1);
104     display.setTextColor(WHITE);
105
106     display.setCursor(10, 10);
107     display.print("TDS: ");
108     display.print(tds_value);
109     display.print(" PPM");
110
111     display.setCursor(10, 30);
112     display.print("EC: ");
113     display.print(ec_Val, 2);
114
115     display.setCursor(10, 50);
116     display.print("Temp: ");
117     display.print(wTemp, 2);
118     display.print(" C");
119
120     // Display water level
121     display.setCursor(10, 60);
122     display.print("Water Level: ");
123     display.print(water_level); // Show raw water level value (0-4095)
124     display.display();
125
126     delay(1500); // Display update delay
127
128     // Send data to ThingSpeak
129     if (client.connect(server, 80)) {
130         String postStr = apiKey;
131         postStr += "&field1=";
132         postStr += String(wTemp, 2);
133         postStr += "&field2=";
134         postStr += String(tds_value);
135         postStr += "&field3=";
136         postStr += String(water_level); // Send water level to ThingSpeak
137         postStr += "\r\n\r\n";
138
139         // Send HTTP POST request
140         client.print("POST /update HTTP/1.1\n");
141         client.print("Host: api.thingspeak.com\n");

```

```
142     client.print("Connection: close\n");
143     client.print("X-THINGSPEAKAPIKEY: " + apiKey + "\n");
144     client.print("Content-Type: application/x-www-form-urlencoded\n");
145     client.print("Content-Length: ");
146     client.print(postStr.length());
147     client.print("\n\n");
148     client.print(postStr);
149
150     delay(500); // Wait for a moment before stopping the connection
151 }
152 client.stop();
153 }
```

References

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